WEEK 2-REPORT

Data Collection, Cleaning, and Exploratory Analysis

Topic : Analysis of Digital Governance Adoption using E-Panchayat Data

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Date: 11/05/26- 17/05/26

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Internship Course Name: Virtual Data Analysis Intern – E-Governance & Digital Services.

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1.Introduction

The main objective of Week 2 was to collect, pre-process, validate, and analyze governance-related datasets associated with the E-Panchayat and e-Gram Swaraj systems in India. The focus of the analysis was to understand the adoption of digital governance practices across different states using Panchayat-level indicators.

Through this dataset, the analysis helps provide an overview of how digital technologies and online governance systems are being utilized in rural areas. The pre-processing and analytical steps were performed using Python and pandas in a Jupyter Notebook environment to prepare the dataset for further exploratory analysis and visualization.

2. Dataset Description.

The dataset used for this analysis contains state-wise governance information related to the E-Panchayat and e-Gram Swaraj systems in India. It includes important indicators such as:

- Gram Panchayat Development Plans Uploaded
- Total Number of Village Panchayats
- Total Number of Block Panchayats
- Zilla Panchayat Plans Uploaded
- Panchayats onboarded to digital systems
- Online payment integration

By comparing the uploaded Panchayat plans with the total number of Panchayats, the dataset helps in understanding the adoption of digital governance technologies across different states. The analysis shows that many Panchayat institutions are gradually moving towards digital platforms and online governance systems.

These insights can be useful for both governmental analysis and public understanding of digital governance implementation in rural areas.

3.Objectives of the Analysis

The main objectives of this analysis were:

1) to collect and study governance-related datasets associated with E-Panchayat and e-Gram Swaraj systems.

2) to pre-process and clean the dataset for accurate analysis.

3) to validate the consistency and reliability of governance indicators.

4) to calculate governance adoption metrics using Panchayat-level data.

5) to analyze the adoption of digital governance systems across different states.

6) to identify variations in digital governance implementation in rural administration.

The analysis also aimed to understand how digital technologies are being integrated into Panchayat governance systems and how these insights can support future governance-related decision making.

4.Tools and Technologies Used

The pre-processing and analysis of the dataset were performed using Python in a Jupyter Notebook environment. The pandas library was mainly used for data cleaning, validation, feature engineering, and exploratory data analysis.

The following tools and technologies were used during the task:

1) Python
2) pandas
3) Jupyter Notebook / Google Colab/VS CODE.
4) CSV datasets
5) Exploratory Data Analysis (EDA) techniques

These tools helped in organizing, cleaning, and analyzing the governance dataset efficiently.

5.Data Collection Process

The dataset used for this analysis was collected from publicly available government governance platforms related to E-Panchayat and e-Gram Swaraj systems in India. The data mainly focused on Panchayat-level digital governance indicators across different states.

The collected dataset include information related to:

1) Gram Panchayat Development Plans Uploaded
2) Block Panchayat Plans Uploaded
3) Zilla Panchayat Plans Uploaded
4) Panchayats onboarded to digital systems
5) online payment integration
6) total Panchayat counts

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The dataset was downloaded in CSV format and imported into a Jupyter Notebook environment using Python and pandas for pre-processing and analysis.

The selection of this dataset was based on its relevance to digital governance adoption, rural administrative digitization, and public governance transparency. The dataset was considered suitable for analysis because it provided measurable governance indicators that could be compared across states.

6.Data Pre-Processing

Several pre-processing techniques were applied to prepare the dataset for analysis. Initially, the dataset structure and column information were inspected using pandas functions such as head() and info().

The pre-processing steps included:

1. Checking missing values
2. Cleaning and organizing column names
3. Standardizing state names
4. Validating duplicate records
5. Converting numerical columns into proper datatypes.

Additional feature engineering was also performed to calculate:

1) total plans uploaded

2) total Panchayat counts

3) adoption percentage of digital governance systems

These pre-processing steps improved the quality and consistency of the dataset before analysis.

7.Data Standardization

Data standardization was performed to maintain consistency in the dataset and improve analytical accuracy. The state names were cleaned and standardized by removing unnecessary spaces and formatting inconsistencies.

Standardization techniques such as string trimming and title formatting were applied to ensure that all categorical values followed a uniform structure. This step helped prevent duplication and inconsistencies during sorting, filtering, and comparative analysis.

8.Missing Value Analysis

Missing value analysis was conducted to identify incomplete or unavailable data within the dataset. The isnull().sum() function in pandas was used to inspect missing values across all governance indicators.

During the pre-processing stage, it was observed that some derived analytical columns initially contained missing values because they had not yet been calculated. After feature engineering and pre-processing, the required governance indicators were successfully generated and validated for further analysis.

9.Feature Engineering

Feature engineering was performed to create additional governance indicators from the existing dataset. New analytical columns were generated to improve comparative analysis and measure digital governance adoption more effectively.

The following derived features were created:

- Total Plans Uploaded

- Total Number of Panchayats

- Adoption Percentage of Digital Governance Systems

The adoption percentage was calculated by comparing the total uploaded Panchayat plans with the total number of Panchayats across states. These engineered features helped in ranking and analyzing governance adoption levels more efficiently.

10.Data Validation

Validation techniques were used to ensure that the dataset was logically correct and suitable for analysis. Several checks were performed to identify possible inconsistencies in the governance indicators.

The validation process included:

- checking duplicate rows
- validating missing values
- identifying negative numerical values
- verifying adoption percentage calculations
- checking datatype consistency

No duplicate records, negative values, or logically impossible adoption percentages were found in the dataset. This confirmed that the dataset was reliable for exploratory analysis and future visualization tasks.

11.Exploratory Data Analysis (EDA)

Exploratory Data Analysis was performed to identify trends and patterns related to digital governance adoption across Indian states.

The analysis included:

1) ranking states based on adoption percentage
2) identifying high-performing and low-performing states
3) generating statistical summaries
4) comparing governance indicators across states

The analysis showed noticeable variation in digital governance adoption levels among different states. Some states demonstrated higher adoption percentages, indicating stronger implementation of online Panchayat governance systems.

12.Preliminary Findings

After cleaning and analyzing the E-Panchayat and e-Gram Swaraj dataset, several important observations were identified regarding digital governance adoption across Indian states.

1. The adoption percentage of digital governance services differs significantly from state to state. Some states show high participation in uploading Panchayat development plans, while others have lower adoption levels.

2. States with higher adoption percentages appear to have better implementation of digital governance systems and stronger usage of online Panchayat platforms.

3. The dataset indicates that digital governance in rural administration is growing, especially through online planning and Panchayat management systems.

4. Some states still show comparatively lower digital adoption, which may be due to differences in infrastructure, awareness, administrative efficiency, or technology accessibility.

5. During pre-processing and validation, no duplicate records, negative values, or logically incorrect adoption percentages were found. This suggests that the dataset is reliable for further analysis and visualization.

6. The calculated adoption percentage helped in comparing governance performance across states and identifying leading and lagging regions in digital governance implementation.

Overall, the dataset provides meaningful insights into the progress of Panchayat-level digital governance in India and forms a strong foundation for further visualization and policy analysis.

13.Conclusion

The Week 2 task successfully involved the collection, pre-processing, validation, and analysis of governance-related datasets associated with E-Panchayat and e-Gram Swaraj systems.

Through pre-processing and exploratory analysis, meaningful governance indicators were generated from the raw dataset. The analysis helped in understanding the adoption of digital governance technologies in Panchayat systems across different states.

The cleaned and validated dataset now provides a good foundation for future visualization, reporting, and policy-related analysis in the upcoming stages of the project.

14.Screenshots and Workflow Documentation

The following screenshots were captured during the pre-processing and analysis workflow:

-Dataset preview using head()

-Dataset structure inspection using info()

-Missing value analysis

-Feature engineering calculations

-Adoption percentage calculation

-Validation checks

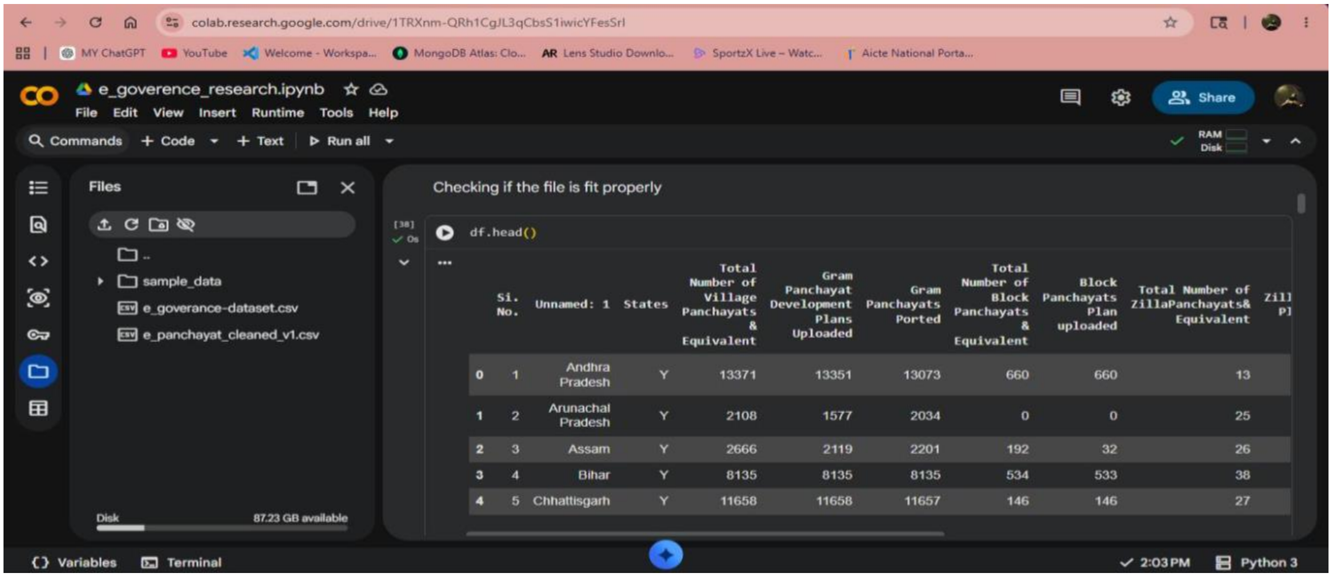
-Ranking analysis of states

-Statistical summary generation

These screenshots provide evidence of the pre-processing, validation, and analytical steps performed during the task.

Figure 1: Dataset Preview.

In Google Colab:



In VS Code:

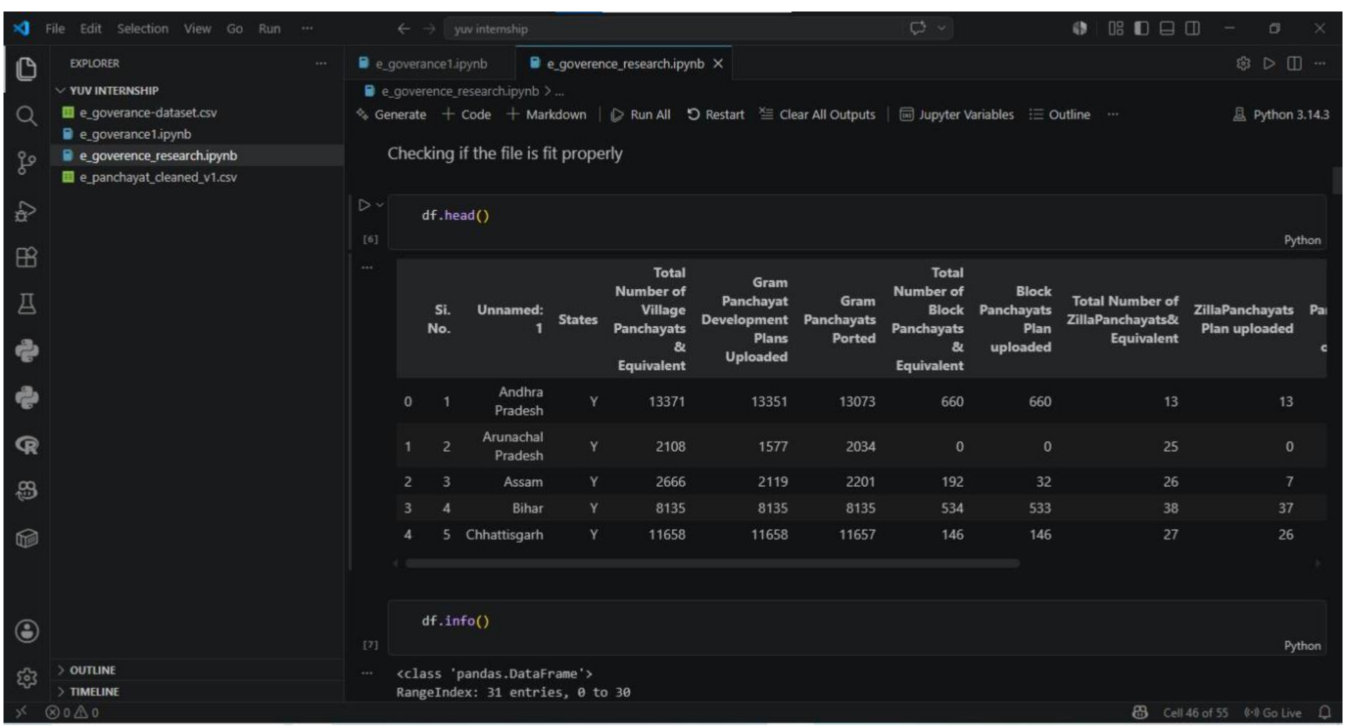
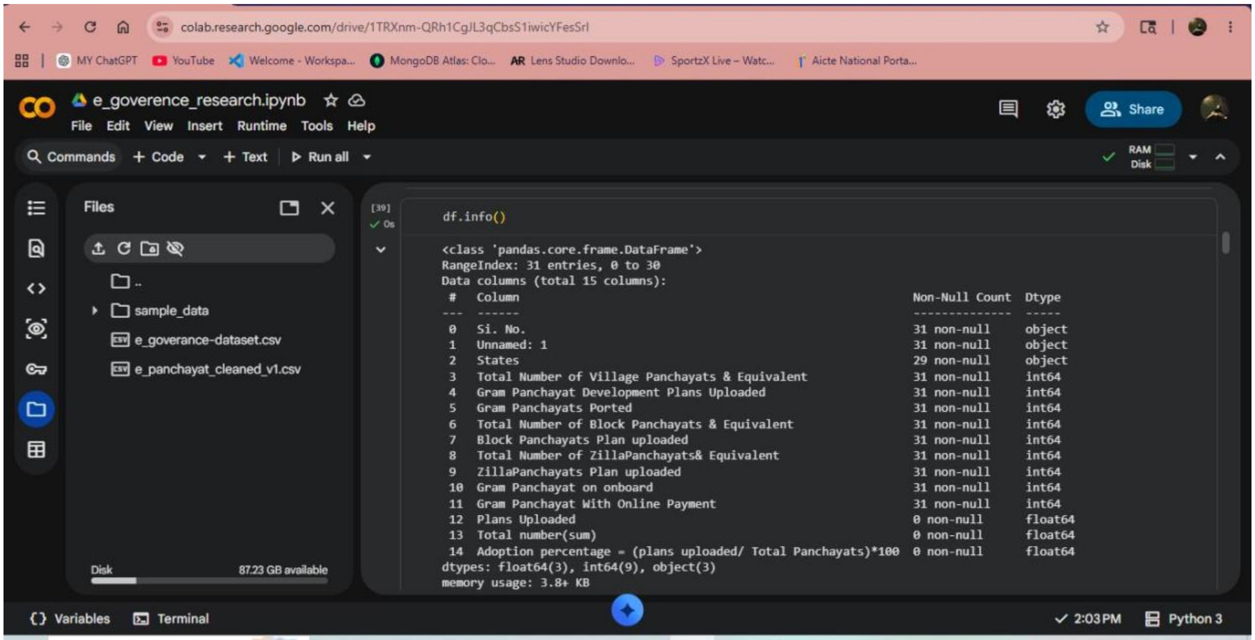


Figure 2: Dataset structure.

In Google Colab :



In VS CODE:

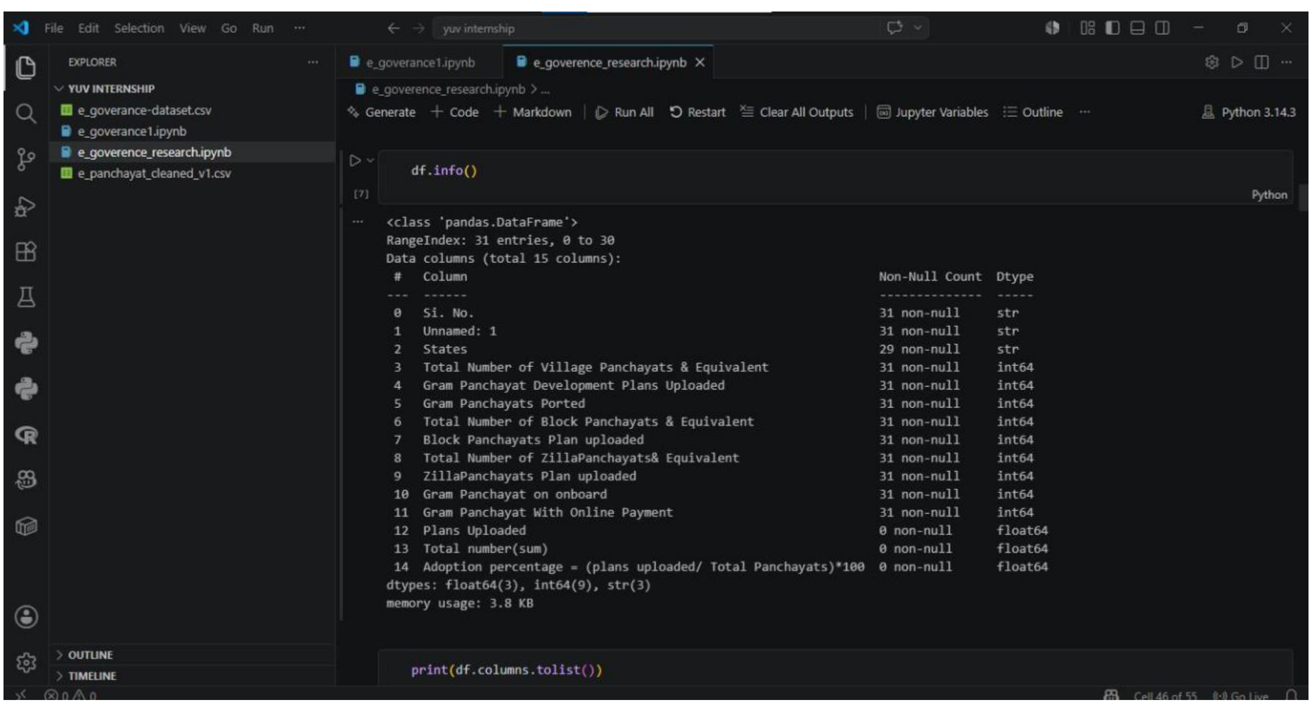
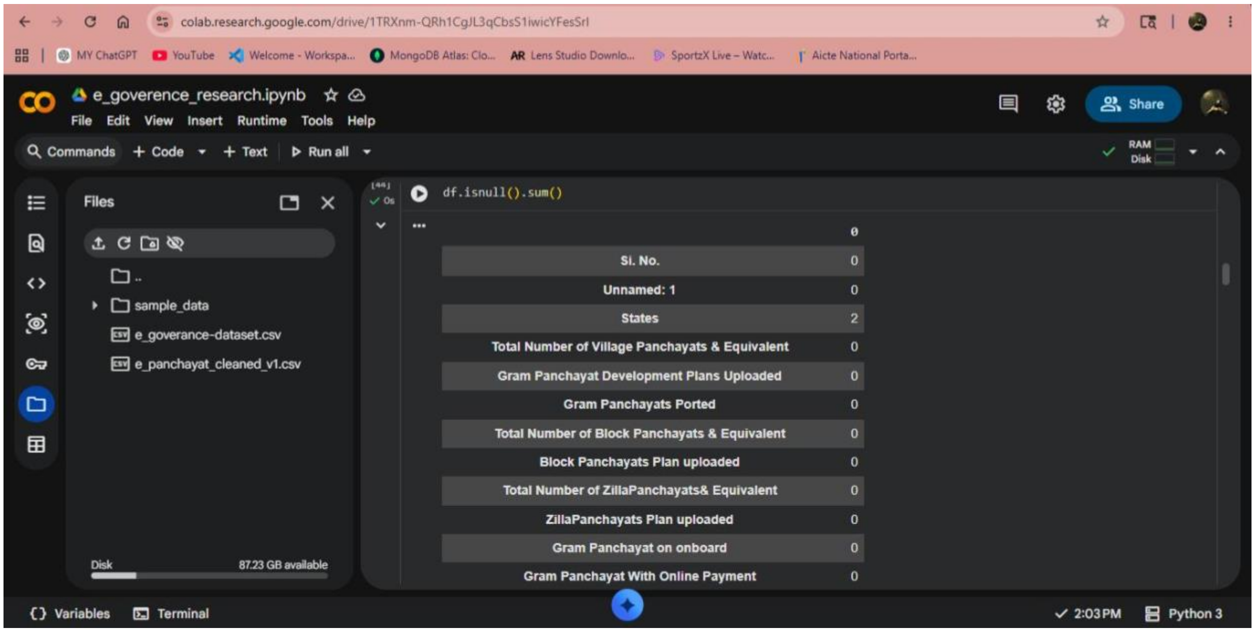


Figure 3: Missing Value Analysis.

In Google Colab:



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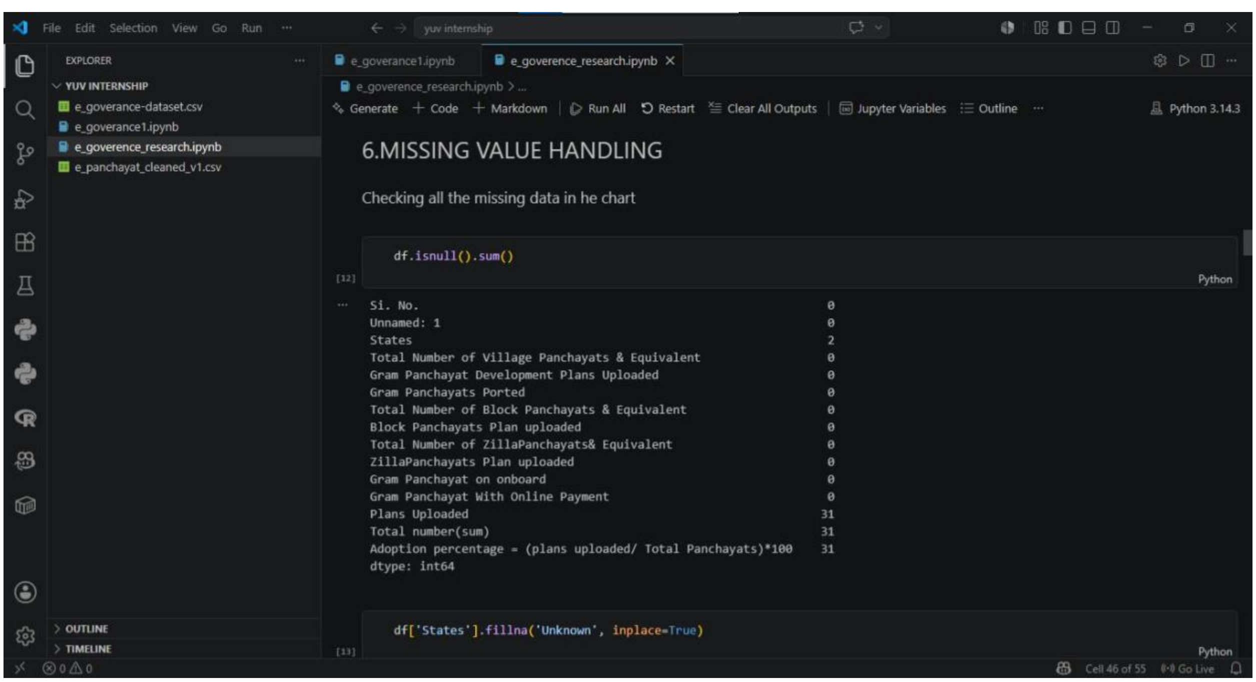
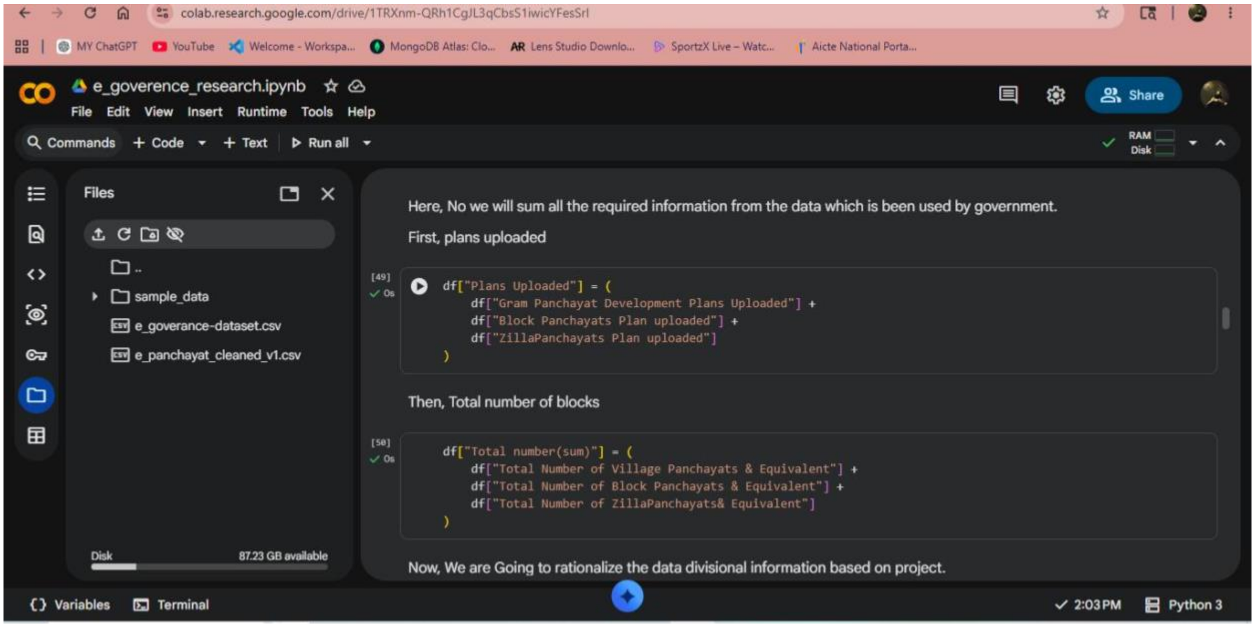


Figure 4: Feature Engineering

In Google Colab:



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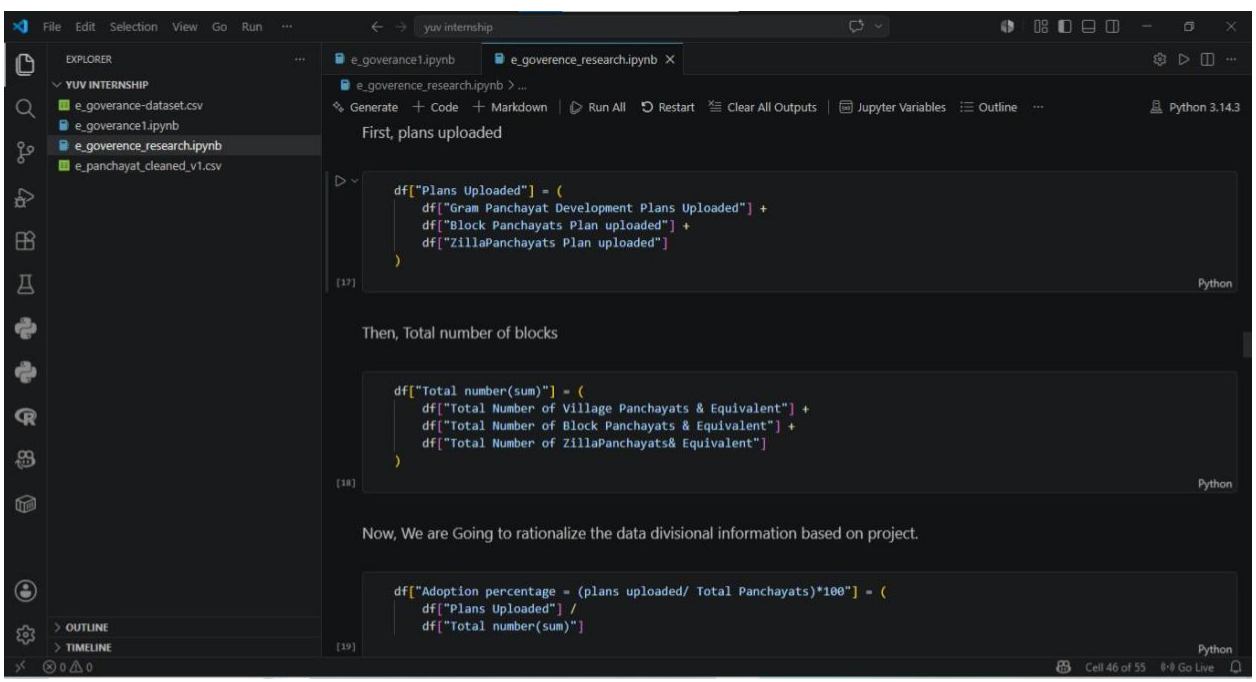
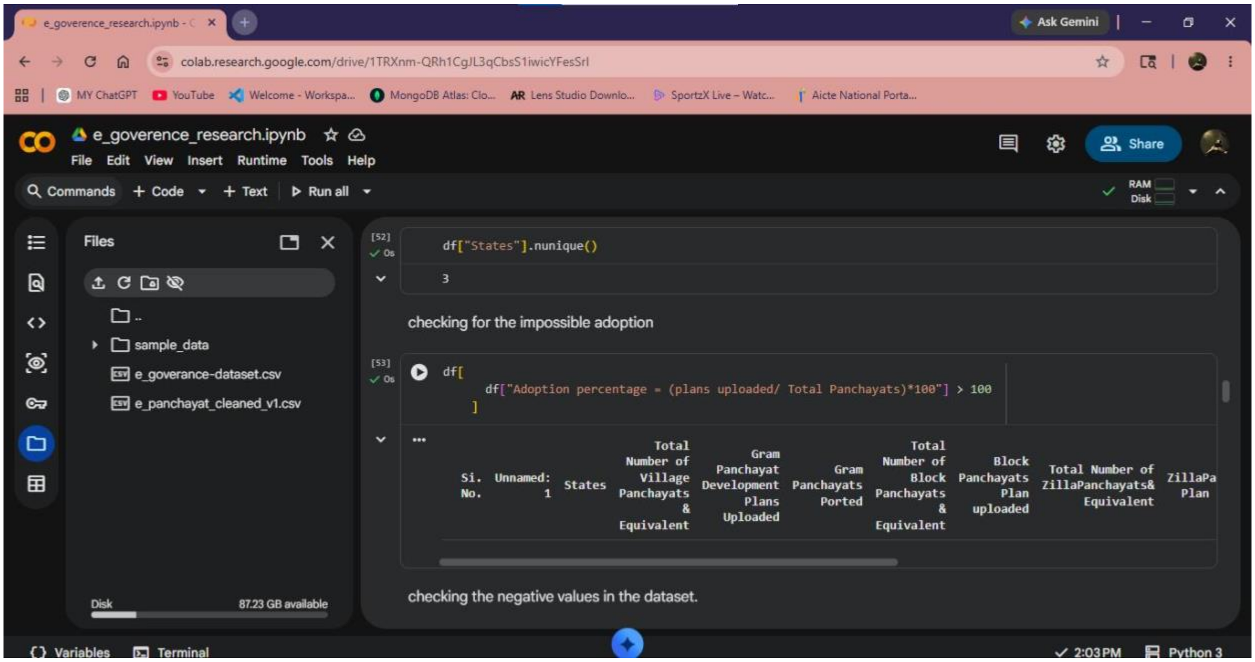
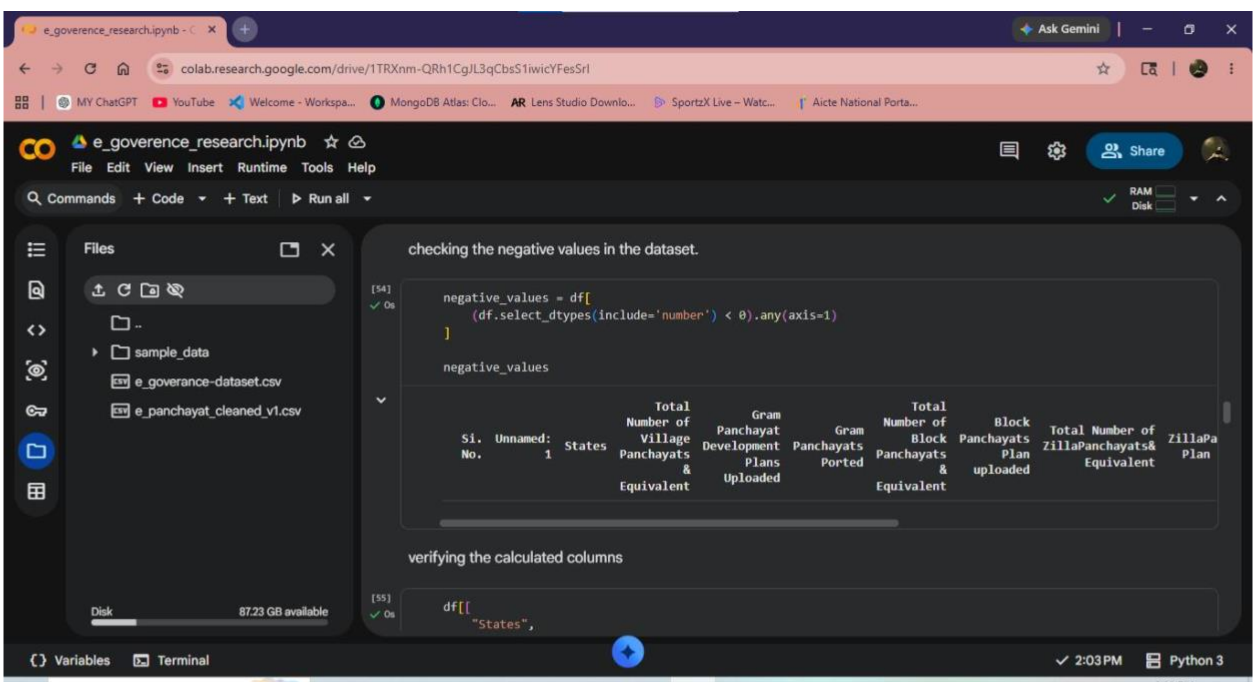


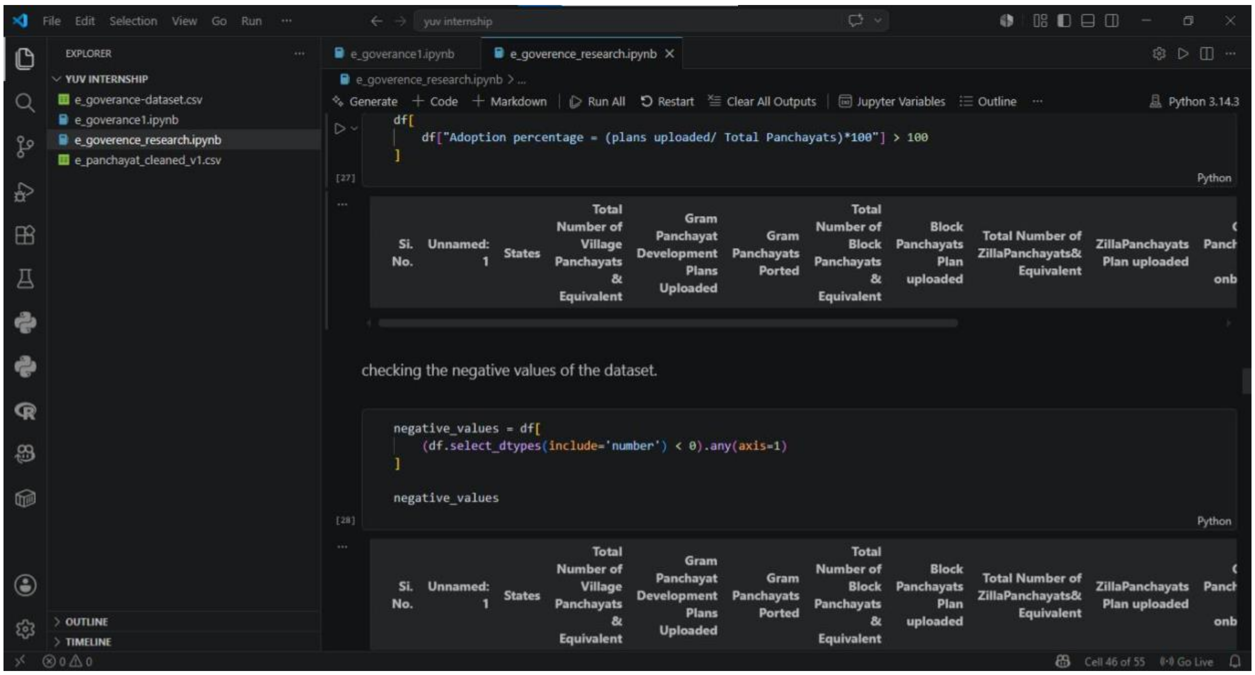
Figure 5: Validation Checks

In Google Colab:





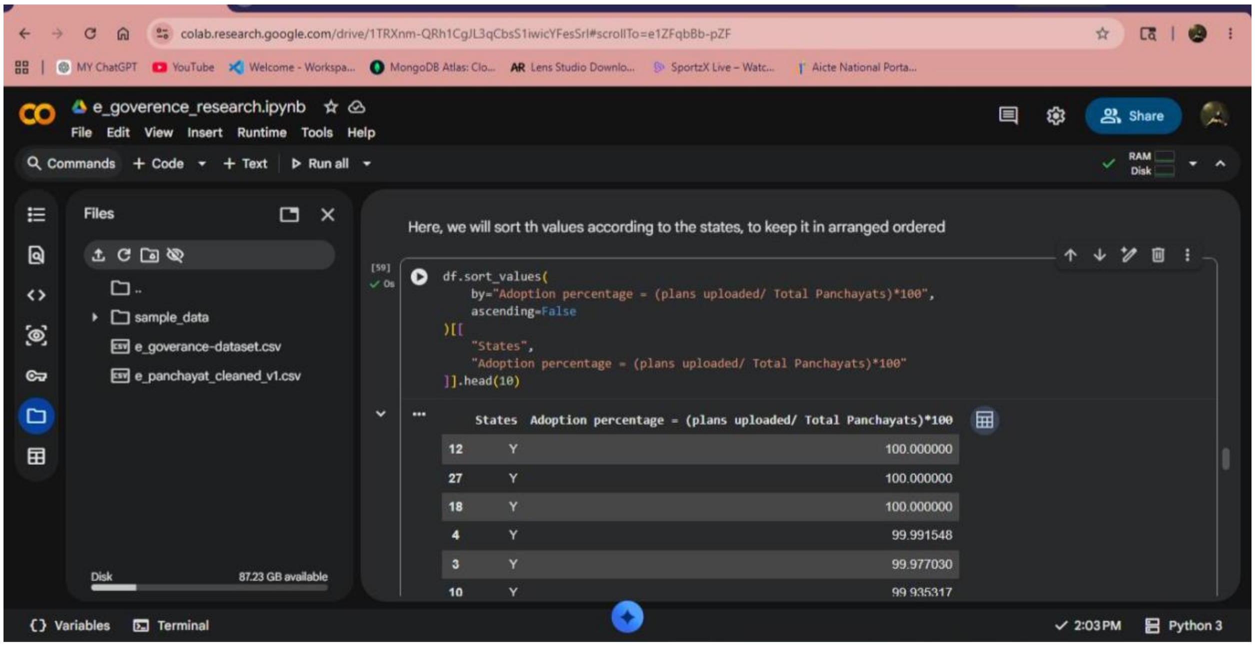
In VS CODE:

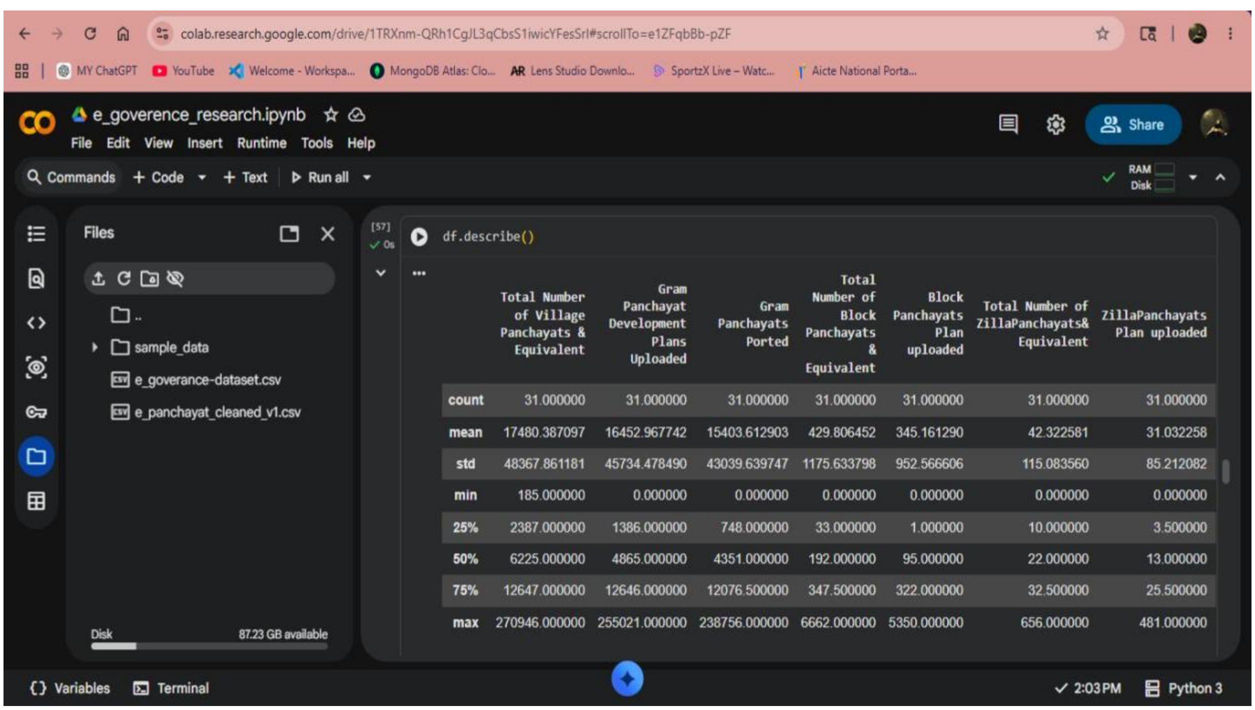


(Scroll below)

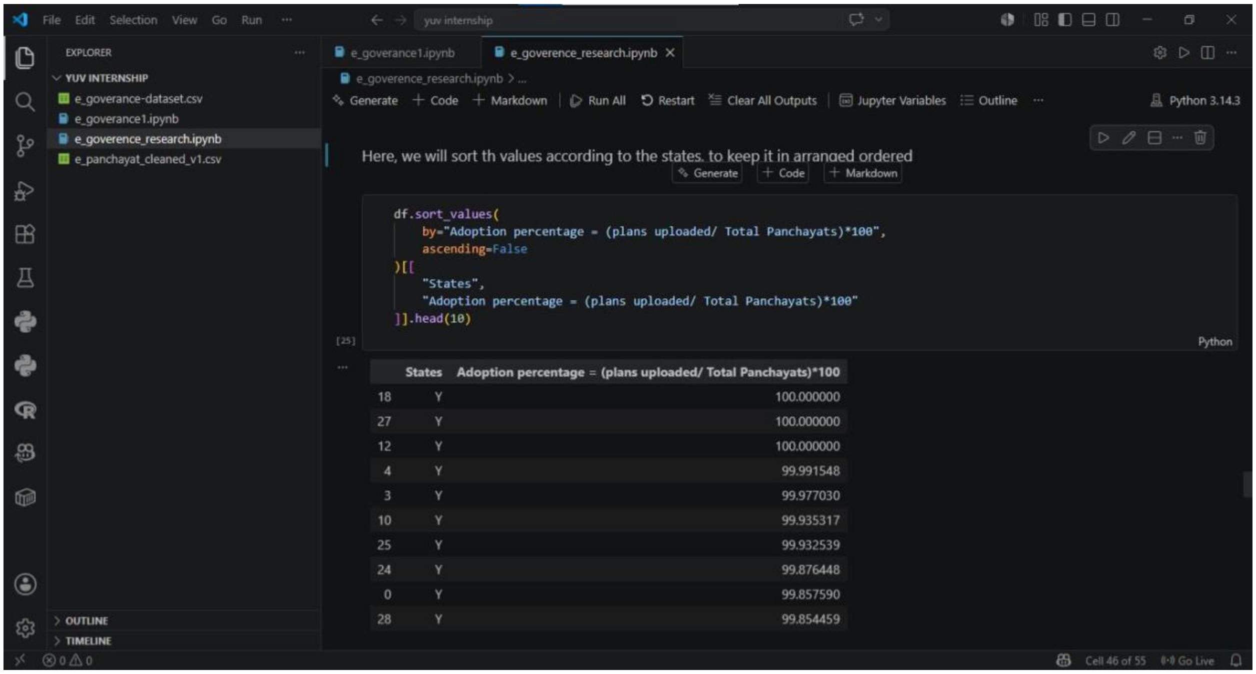
Figure 5: Ranking Analysis

In Google Colab:





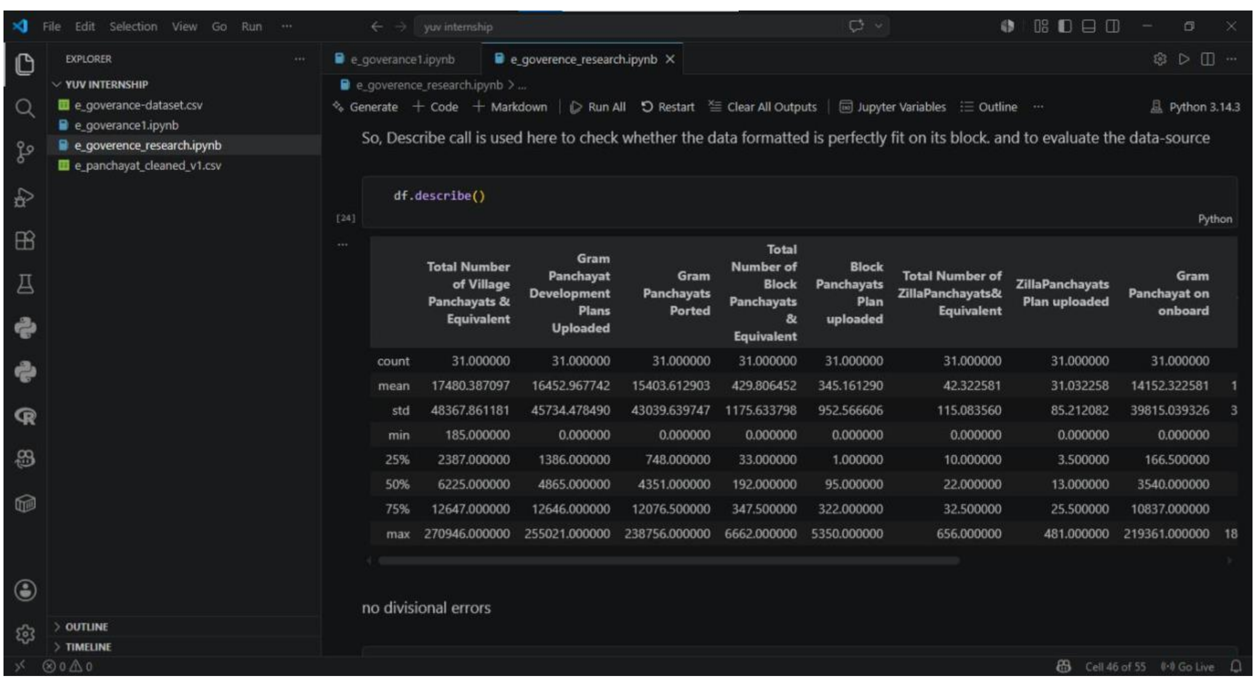
In VS CODE:



Here, we will sort th values according to the states. to keep it in arranged ordered

```python
df.sort_values(
    by="Adoption percentage = (plans uploaded/ Total Panchayats)*100",
    ascending=False
)[[
    "States",
    "Adoption percentage = (plans uploaded/ Total Panchayats)*100"
]].head(10)
```

	States	Adoption percentage = (plans uploaded/ Total Panchayats)*100
18	Y	100.000000
27	Y	100.000000
12	Y	100.000000
4	Y	99.991548
3	Y	99.977030
10	Y	99.935317
25	Y	99.932539
24	Y	99.876448
0	Y	99.857590
28	Y	99.854459



So, Describe call is used here to check whether the data formatted is perfectly fit on its block. and to evaluate the data-source

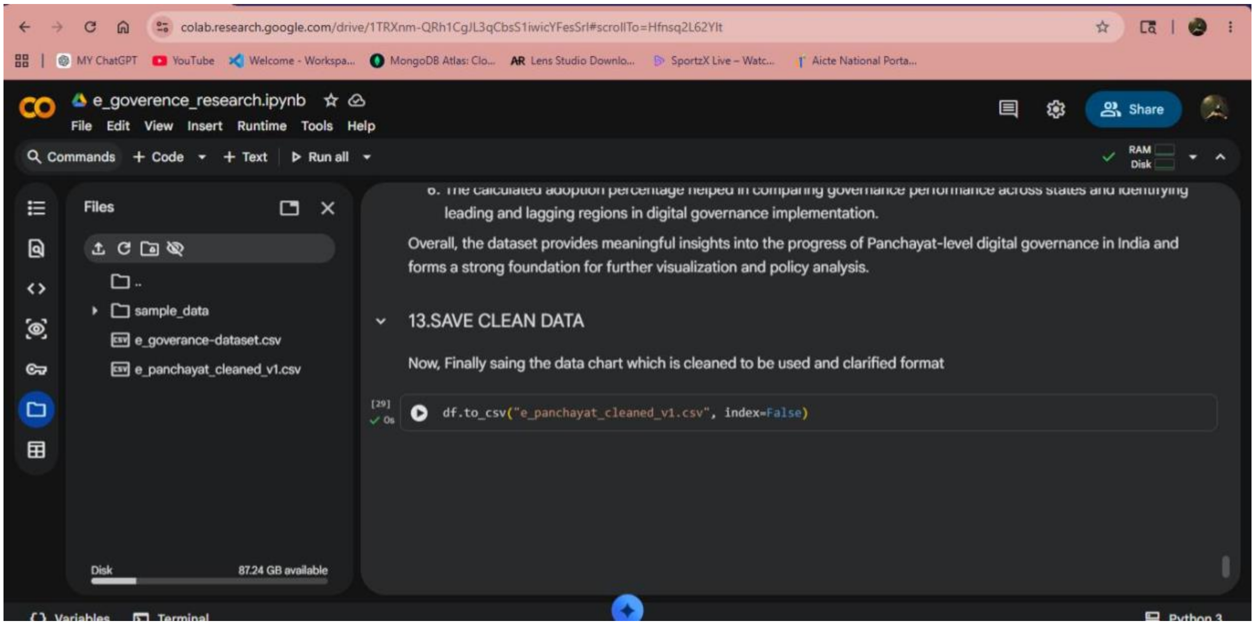
```python
df.describe()
```

	Total Number of Village Panchayats & Equivalent	Gram Panchayat Development Plans Uploaded	Gram Panchayats Ported	Total Number of Block Panchayats & Equivalent	Block Panchayats Plan uploaded	Total Number of ZillaPanchayats& Equivalent	ZillaPanchayats Plan uploaded	Gram Panchayat on onboard	
count	31.000000	31.000000	31.000000	31.000000	31.000000	31.000000	31.000000	31.000000	
mean	17480.387097	16452.967742	15403.612903	429.806452	345.161290	42.322581	31.032258	14152.322581	1
std	48367.861181	45734.478490	43039.639747	1175.633798	952.566606	115.083560	85.212082	39815.039326	3
min	185.000000	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	
25%	2387.000000	1386.000000	748.000000	33.000000	1.000000	10.000000	3.500000	166.500000	
50%	6225.000000	4865.000000	4351.000000	192.000000	95.000000	22.000000	13.000000	3540.000000	
75%	12647.000000	12646.000000	12076.500000	347.500000	322.000000	32.500000	25.500000	10837.000000	
max	270946.000000	255021.000000	238756.000000	6662.000000	5350.000000	656.000000	481.000000	219361.000000	18

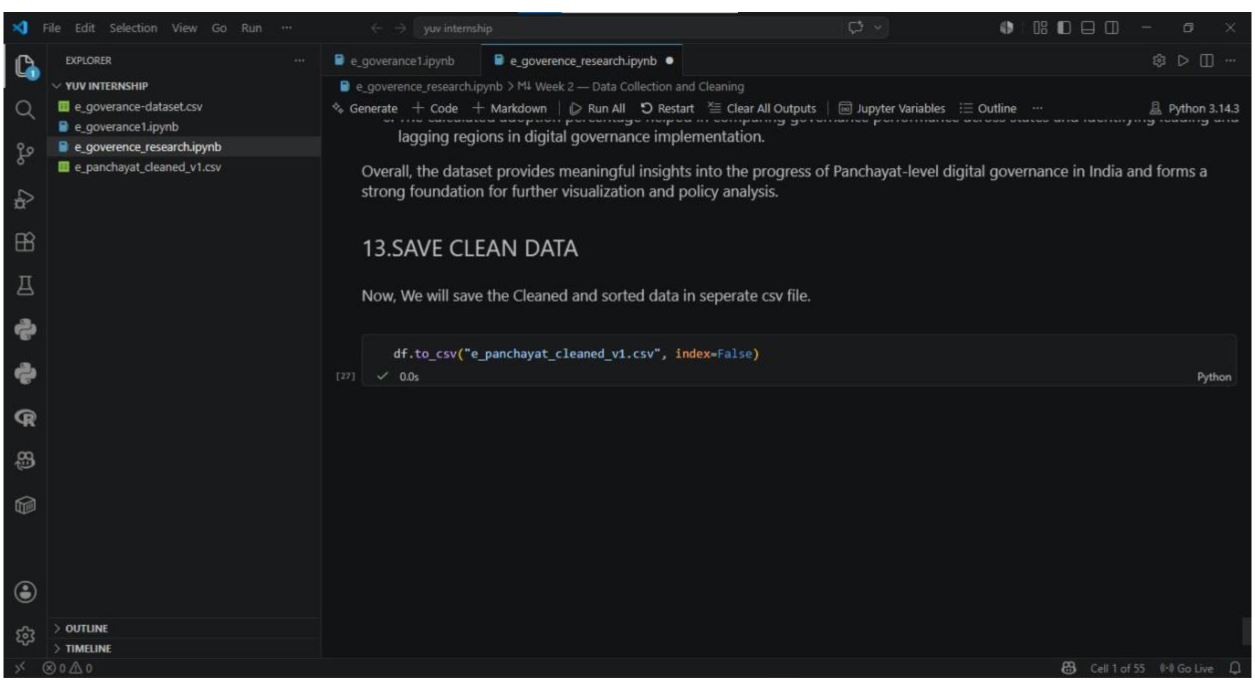
no divisional errors

Figure 6: Statistical summary generation.

In Google Colab:



In VS CODE:



15.References

1. E-Gram Swaraj Portal – Government of India

https://egramswaraj.gov.in/

2. Ministry of Panchayati Raj – Government of India

https://panchayat.gov.in/

3. Python Documentation

https://www.python.org/

4. Pandas Documentation

https://pandas.pydata.org/docs/

5. Jupyter Notebook Documentation

https://jupyter.org/

6. India Open Government data:

https://www.data.gov.in/

7. GitHub – E governance & Digital Service:

https://github.com/rohanw2004/E--Governance-Digital-Service